# World of Quantum Materials

A player's guide

## Contents



## The Game

**You are a quantum material.** A browser-based role-playing game that teaches Aalto University's *Advanced Quantum Materials* course from the inside: you start as Silicon, walk ten worlds (one per course topic), and battle the real compounds that ambush you, using only the quasiparticles your own lattice can host. Every move is a genuine excitation (magnons, phonons, spinons, Majorana modes), and it lands at **double damage** on a material whose physics cannot host it, so winning means knowing which phase you are facing. Meanwhile a "Decoherence" is stripping the worlds of their protected properties, and you are the one walking each phase of matter back into shape.


Figure 1: Title screen

**Play it**

**In your browser**, nothing to install:

**https://joselado.github.io/world_of_quantum_materials/**

Works on any current browser on Windows, macOS or Linux. Needs no account, no download and no permission to install software, so it runs on a locked-down university machine as happily as on your own laptop. You need to be online to open it.

**Offline, as a single file.** Download `world_of_quantum_materials.html` (right-click the link, "Save Link As…") and open it by double-clicking it. The whole game is inside that one file, because every sprite and every note is drawn and played by code rather than loaded from anywhere, so once it is on your machine it never touches the network again.

**From the source**, if you want to read or change the game. Install Node.js 18+, then from the repo root:

```
npm run play
```

That installs what it needs on first run and opens the game. Same command on Windows, macOS and Linux.

Your progress is saved in the browser you play in (or, for `world_of_quantum_materials.html`, in the browser you open it with), so it stays on that machine, and clearing your browsing data clears it. There are no accounts and nothing is sent anywhere.

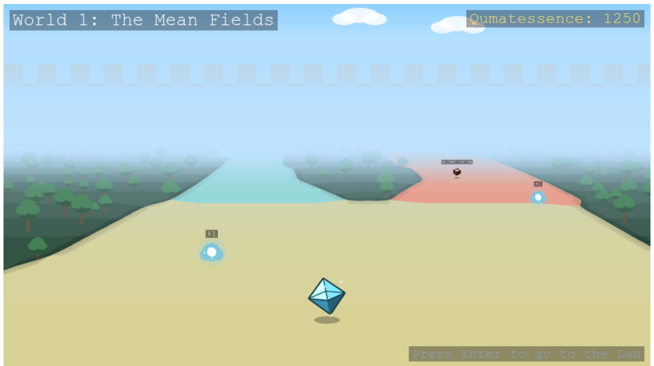
**The premise**

**You are a quantum material yourself**, the same kind of matter the wild encounters are drawn from, starting out as Silicon. A "Decoherence" is spreading through these worlds, causing them to lose their protected properties, and you're the one walking through each phase of matter to stabilize it again.

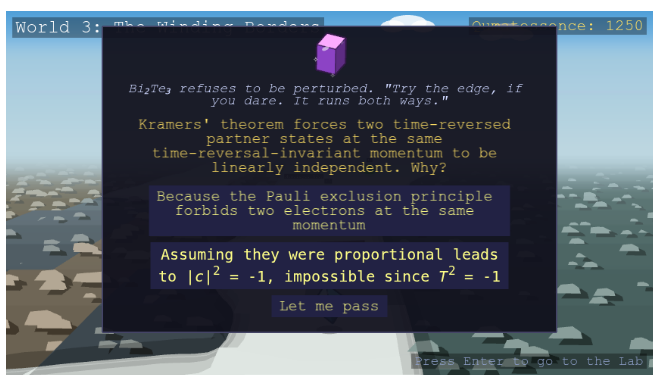
**How it plays**

**Explore.** Each world is walkable ground rendered in an over-the-shoulder pseudo-3D view, the world redrawn from a smoothly moving camera as you walk. Every world's layout echoes its own physics, not just its scenery: a corridor that splits into two colored branches and remerges in the mean-field world, an open cloister you cross between rows of columns in the tight-binding world, a network of colored domains you trace the boundary of in the topological world, a spiral around a frozen vortex in the superconductivity world, a real ladder of linked lanes in the entanglement world. Reaching the far end means reading that shape, not just holding one direction. Pickups of qumatessence (the in-game currency) are tucked along the way, often at a dead end worth the detour, and a world you have picked clean quietly refills as you walk it, always out of sight.

The ten worlds are one road, and the light dies along it, from morning through storm and night to no sky at all, after which every world lights itself. Stepping off the path is never safe, and what it costs you gets worse with every world. From each world's far end you can see the next one on the horizon.



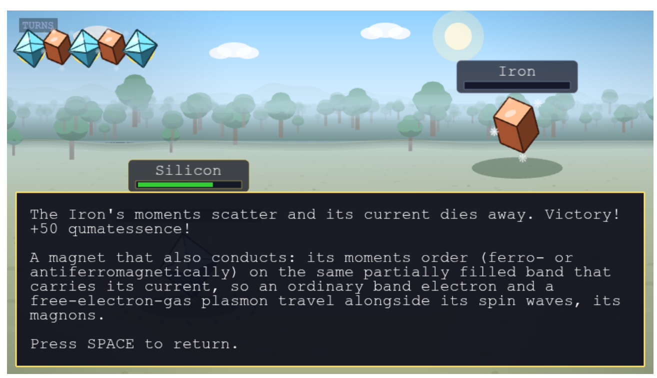

**Get ambushed.** Bump into a wild crystal and it starts a conversation, not just a fight. Most ask a short physics question pulled straight from the course material: answer correctly and you go into battle with a power boost, answer wrong and you're weakened, or just say "let me pass" and walk on with no consequence either way. Every compound has its own look too, drawn in the crystal habit its real lattice grows in (a cubic compound is a blocky cube, a monolayer a thin sheet floating over its own shadow), with its own tilt, proportions and tint, so a wild Chromium reads as its own crystal standing next to the Nickel Oxide beside it.





**Battle.** Turn-based, speed-ordered by your crystal's Momentum stat, and the faster side doesn't just go first, it swings more than once a round if it's fast enough. Every swing is its own pick, so a fast crystal chooses a fresh move for each one. Every move is a real quasiparticle, and every material can only ever learn the moves its actual physics supports: a plain band insulator never gets a magnon move, since it has no magnetic order to carry one. If a defender's own physics can't host your move's quasiparticle at all, it lands at **double damage**, the one type-interaction rule in battle. See Quasiparticles & moves for the full move list and which crystal types can use each one.

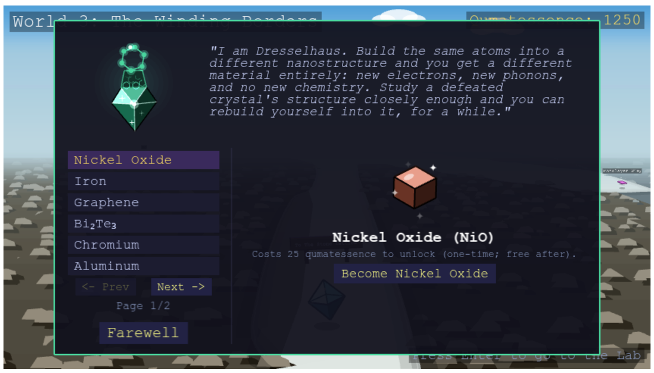



**Grow.** Winning battles and grabbing qumatessence pickups funds the guardians waiting partway through each world, ten in all, each teaching a different way of bending the game's usual rules: new moves and stats, teleportation, transmuting into a crystal you've defeated, always-on passive abilities, fusing two crystals into a hybrid, quiz-gated power moves, and a capstone quiz-gated ultimate move. Each one greets you the first time you reach their row, and once you have met them you can walk back up to them any time and press Space to talk again. Every guardian you've met also stands in the Lab, one click away. See Guardians for what each one does.

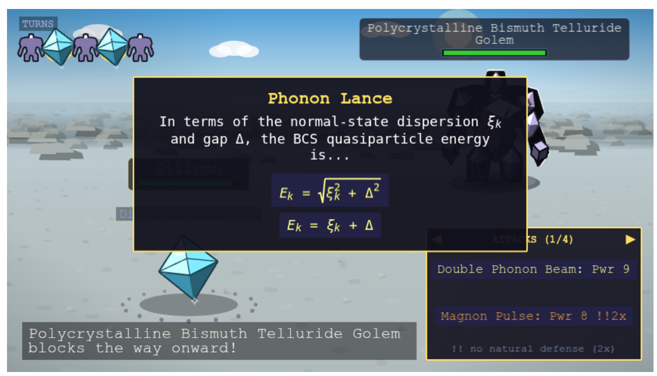

World 1: The Mean Fields
Qumatessence: 1250
"I am Noether. Every symmetry hides a conservation law. Spend your qumatessence on a new attack, or a sharper stat."
v MOVES
Electron Pulse
Phonon Beam
> STATS
Farewell
Electron Pulse
Costs 35 qumatessence.
Learn Electron Pulse
Press Enter to go to the Lab


World 2: The Stone Lattice
Qumatessence: 1250
"I am Bloch. Name a world you have already touched, and I will fold you there."
The Mean Fields
The Stone Latti…
The Winding Bor…
The Storm Flats
The Vortex Glac…
The Iron Steppe
<- Prev
Next ->
Page 1/2
Farewell
Left alone, the field's spins have no reason to point anywhere: the Hamiltonian doesn't care. Then one mean-field interaction tips the balance, and the whole system commits to a single broken symmetry it can no longer see a reason to abandon.
Costs 15 qumatessence to unlock (one-time; free after).
Travel to The Mean Fields


World 3: The Winding Borders
Qumatessence: 1250
"I am Dresselhaus. Build the same atoms into a different nanostructure and you get a different material entirely: new electrons, new phonons, and no new chemistry. Study a defeated crystal's structure closely enough and you can rebuild yourself into it, for a while."
Nickel Oxide
Iron
Graphene
Bi₂Te₃
Chromium
Aluminum
<- Prev
Next ->
Page 1/2
Farewell
Nickel Oxide (NiO)
Costs 25 qumatessence to unlock (one-time; free after).
Become Nickel Oxide
Press Enter to go to the Lab


World 5: The Vortex Glacier
Qumatessence: 1250
"I am Majorana. Fuse two states you understand and see what phase they make together."
Cr-doped (Bi,Sb…
Fe/Pb Majorana …
HgTe/CdTe Quant…
InAs/Al Majoran…
Twisted Bilayer…
Farewell
HgTe + CdTe
HgTe/CdTe Quantum Well
A thin inverted layer between ordinary barriers hides a protected edge neither parent had.
Costs 60 qumatessence to unlock (one-time; free after).
Fuse
Press Enter to go to the Lab


World 4: The Storm Flats
Qumatessence: 1250
"Put a strong enough field on a two-dimensional electron gas and its whole band breaks into a ladder of flat levels, one fixed quantum of energy apart. Tell me the physics right and I will teach your crystal to strike by that ladder. Answer right and the hit climbs a rung and lands twice as hard. Answer wrong and it barely lands at all. Tell me which quasiparticle should carry it, too."
v LANCE
Electron
Phonon (curren…
> ERUPTION
Farewell
Phonon Lance
Already tuned to Phonon.
Press Enter to go to the Lab


TURNS
Polycrystalline Bismuth Telluride Golem
Phonon Lance
In terms of the normal-state dispersion $\xi_k$ and gap $\Delta$, the BCS quasiparticle energy is...
$E_k = \sqrt{\xi_k^2 + \Delta^2}$
$E_k = \xi_k + \Delta$
ATTACKS (1/4)
Double Phonon Beam: Pwr 9
Magnon Pulse: Pwr 8 !!2x
!! no natural defense (2x)
Polycrystalline Bismuth Telluride Golem blocks the way onward!


World 10: The Devouring Mirror
Qumatessence: 1250
"I am Skłodowska-Curie, and I lead this circle of guardians: Noether, Bloch, Dresselhaus, Landau, Majorana, Anderson, Feynman, Kondo, Franklin, and I. Here is our last lesson. Answer three questions on the physics running through everything you have learned. Get all three right and your crystal strikes with a force none of the others can match. Miss even one and the blow lands nowhere at all. Tell me which quasiparticle carries it, too. A new one costs a lot to unlock, but once bought it is yours to wear again for free."
v METEOR
Electron
Phonon
> NOVA
Farewell
Phonon Meteor
Phonon is already yours. Carrying it again is free.
Carry Phonon
Press Enter to go to the Lab

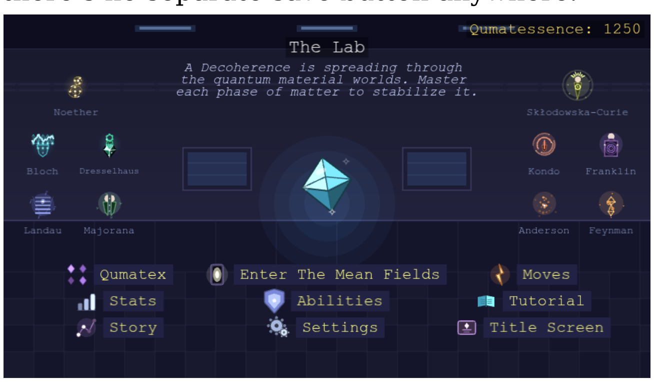

World 7: The Entangled Web
Qumatessence: 1250
"A tensor network and a Feynman diagram draw the same trick two ways: a vertex for every point, a line for every leg. Show me you understand a move you already carry, and I will draw a higher-order correction into it. Paid for whether it lands or not."
Phonon Beam
Spin Screening
Phonon Lance
Phonon Meteor
Farewell
Double Phonon Beam
Swing it at:
Base
Double
Level to "Triple": 4 questions in a row, 60 qumatessence paid whether it lands or not.
Level to "Triple"
Press Enter to go to the Lab

Figure 2: Anderson's impurity-doping panel

**Face the boss.** Every world narrows into a pass at its far end, and you will see what stands in it long before you get there: a gigantic golem, built from many crystal shards fused into one mass, filling the gap so completely that nothing of the next world shows past it. Through World 9 its name is always a real compound in *polycrystalline* form (many grains fused into one, the same idea the golem's own body literalizes). Each one boasts about the very property its own grains took from it, and none of them knows that is what it is doing. World 10's rival, The Adapted, is the one that breaks the pattern. Step to the mouth of the pass and press Space to challenge it. Beat it and the pass clears: the next world's colours show through the gap, a board names it, and pressing Space again crosses over.

**Walk back anytime.** The near end of every world, right where you first walked in, is a pass too, with its own board and nobody guarding it, leading back to the world before it (or the Lab, from World 1). Walk up and press, no menu required; you'll land back in that earlier world right at its own far end, ready to walk forward through it again whenever you like.

**Return to the Lab.** World 0 is a static hub room, each of its jobs its own physical station: Qumatex, a filterable index of every crystal in the game listed by name alongside a note on the real physics behind it, each with a "???" placeholder name and silhouette for anything you haven't found yet; a door back out to the furthest world you've reached; and stations to check your moves, your stats, replay the tutorial, re-read the story so far, and adjust settings. Every guardian you've met stands in the room itself, so you can click one to reopen their panel without leaving. Your progress autosaves as you play, so there's no separate save button anywhere.




### The ten worlds

One world per course topic, each with its own biome, wild-material archetypes, and a rival crystal gating the way to the next world.

| # | World | Course topic |
|---|---|---|
| 1 | The Mean Fields | Second quantization, mean-field, symmetry breaking |

| # | World | Course topic |
|---|---|---|
| 2 | The Stone Lattice | Symmetries, tight-binding band structure |
| 3 | The Winding Borders | Topological band theory |
| 4 | The Storm Flats | Integer and fractional quantum Hall effect |
| 5 | The Vortex Glacier | Superconductivity, Nambu representation, Majoranas |
| 6 | The Iron Steppe | Classical magnetism and magnons |
| 7 | The Entangled Web | Quantum entanglement and tensor networks |
| 8 | The Screened Swamp | Quantum magnetism, spinons, Kondo physics |
| 9 | The Defect Scars | Excitations and defects |
| 10 | The Devouring Mirror | Machine learning for quantum materials |

World 10's wilds are every hybrid crystal fusable at Majorana's station (see Hybrids), real compounds in their own right, just ones reached by fusing two parents rather than found unmixed anywhere else. Its rival, The Adapted, is a final boss built as "a model of you": it starts the fight mirroring whichever type you're currently wearing, then reshapes itself every time you land a hit, transmuting live into a real compound that hosts whatever quasiparticle class you just attacked with. See Crystals for every world's full wild-material list.

Each world has its own history, its own way the Decoherence comes for it, and its own rival standing in the way, and the ten of them tell one story. The story is that story, world by world, from the premise to the ending. **It spoils everything**, including who the final enemy turns out to be, so it's for a second playthrough or for a player who wants their bearings more than the surprise.

### Battle basics

Turn order runs on Momentum: the faster side swings first, and swings again (up to 5 times a round) the more its Momentum outpaces the other side's, while the slower side always still gets its own hit. Each of your swings is a separate choice from the move menu, and the "Turns" row in the top-left corner counts down how many you have left before the other side acts. That applies to the quiz-gated moves too: every swing you spend on one asks its question again. Energy raises your crit ("coherent hit") chance and Lifetime raises your defense; every stat runs from 1 to a cap of 100, raised one point at a time at Noether's shop. HP is fully healed after each battle, so qumatessence, not HP attrition, is what is actually on the line from one fight to the next. The move menu shows one kind of move at a time (ordinary attacks, quiz-gated moves, or status-inflicting moves), with the Left/Right keys or the on-screen arrows to page between kinds.

For the full mechanics, meaning every move and which crystals can use it, hybrid materials, and what each guardian teaches, see:

- Quasiparticles & moves
- Crystals
- Hybrid materials
- Guardians
- The story: spoils the whole plot, including the ending

All of it is also collected in **the player's guide**, a single printable document: this page, the story, and every reference table in one place.

### Controls

Play uses both: the mouse chooses, the keyboard walks. On a phone or a tablet, a finger does both.

**With the mouse.** Everything on screen that offers something is clickable: the Lab's stations and the guardians standing in the room, every panel button, the prompt that appears at a pass, the answers to a wild crystal's question, your moves in battle, the arrows that page a long list or a question too long for one screen, and the line in the bottom right corner of a world that takes you back to the Lab.

**With the keyboard.**

| Key | Action |
|---|---|
| Arrow keys | Walk (Up/Down forward and back, Left/Right sideways) |
| Space | Take whatever is offered where you stand: greet a wild crystal, talk with a guardian you have met, challenge a rival, cross a pass, carry on after a battle |
| Enter | Return to the Lab (World 0) |
| Enter, while in the Lab | Head back out to whichever world you're in the middle of (once you've entered one) |
| Left / Right, in battle | Page between kinds of move |

Picking a move in battle or opening a station in the Lab is the one thing the keyboard can't do.

**On a touchscreen.** Four arrows sit in the bottom left corner of every world: hold one and you walk, exactly as if you were holding the matching key. Tap the line in the bottom right corner to return to the Lab, and tap anywhere to leave the summary after a battle. Everything else was already a tap. The arrows appear on their own when you open the game on a phone or a tablet, and the Settings station can turn them on or off by hand. The game is built for a screen wider than it is tall, so turn the phone sideways.

The Lab is where everything else lives: your moves and stats, tutorial replays, the story so far, and settings, each its own station in the room (see "Return to the Lab" above).

### Settings

In the Lab, click the **Settings** station. The settings come in three groups, Gameplay, Story and Presentation, and you switch between them at the top of the panel. Within a group each setting is a row, with every value it can take laid out beside its name; click a value to pick it, and the one in gold is the one you are playing with.

**Gameplay.**

- **Difficulty**: B.Sc., M.Sc., or Ph.D., how hard every world's opponents hit. Meant to be changed mid playthrough rather than picked once, and it applies to your very next battle.
- **Enemy Density**: Low, Normal, High, or Very High, if wild crystals feel too sparse (or too frequent) along the path. Sets how many crystals a world holds at once, so it also governs how many drift back in after you have fought them. Takes effect the next time you enter or re-enter a world.
- **World Size**: Nano, Meso, or Macro. Every world keeps its own shape and changes how big it is: Macro corridors run three times as wide and three times as far, Nano ones are a brisk run through the same place. Crystals and qumatessence are spread through it at the same rate either way, so a bigger world holds proportionally more of both. Takes effect the next time you enter a world.

**Story.**

- **Story Screens**: On or Off, whether a world's opening lore, a rival's taunt and the beat between worlds stop you as you play. Turning them off loses nothing: everything they would have shown fills into the Lab's Story station just the same, waiting there whenever you want to read it.
- **Tutorial Tips**: On or Off, the popups explaining a feature the first time it comes up. Off, they still fill into the Lab's Tutorial station on exactly the same schedule, so you can read any of them at your own pace. Handy on a second playthrough, or any time you would rather not be stopped.

**Presentation.**

- **Text Size**: Compact, Normal, or Large. Applies immediately to every menu and dialogue in the game.
- **Music**: Classic or Modern, two renditions of every world's soundtrack. Classic is the original chiptune style; Modern plays the same tunes with softer, smoother tones. Or Mute for no music at all. Sound effects keep playing either way. Applies immediately, and is remembered between sessions.
- **Touch Controls**: Auto, On, or Off, the walking arrows in the corner of the world. Auto puts them there when you are playing on a touchscreen and leaves them off otherwise, which is what you

want unless you want the opposite. The keyboard keeps working whichever way it is set. Takes effect the next time you enter a world.

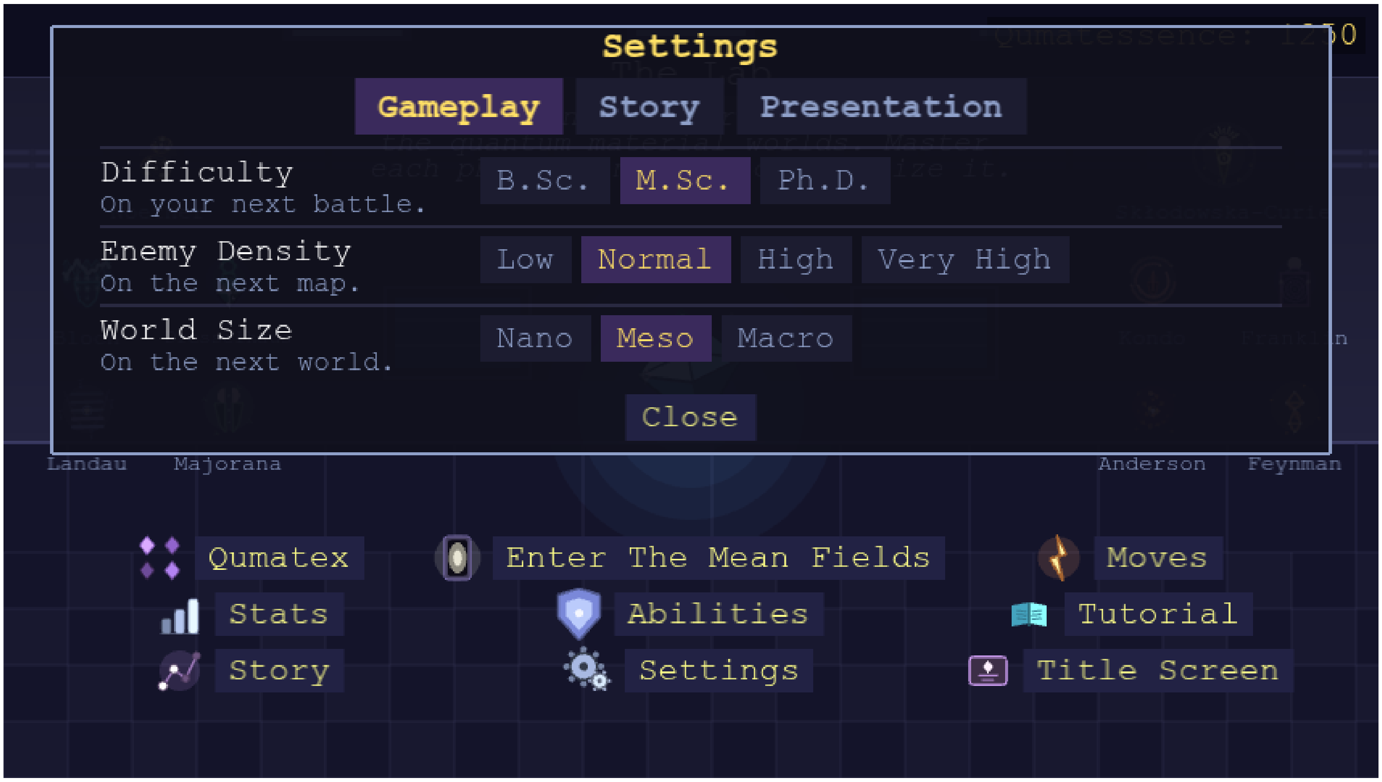


Figure 3: Settings panel

**Tutorial**

Short tips appear one at a time, right as each feature first comes up: entering the Lab, your first steps, your first encounter, your first battle, your first guardian. In the Lab, click **Tutorial** to reread any of them, alongside a topic for every guardian's ability, the Settings station, and Story Mode vs. Superposition Mode. In Story Mode the list fills in as you play; in Superposition Mode every topic is listed from the start. If you would rather not be stopped by the popups at all, the Settings station turns them off, and the list still fills in as you reach each one.

Figure 4: A tutorial tip introducing the guardians

**Story Mode vs. Superposition Mode**

Before you start, the title screen asks you to pick a mode:

- **Story Mode** is the normal playthrough: start at World 1, defeat each world's rival to open the next one, meet each guardian in turn.
- **Superposition Mode** is for players who want the whole game open from the first minute, with no story to work through first. Every guardian is already met and everything they teach is unlocked from the moment you start. Your stats, moves and HP are automatically brought up to each world's level, Bloch's teleport hub offers every world immediately, and Dresselhaus, Majorana and Anderson offer every crystal in the game. Several starting picks are randomized, so a fresh Superposition save rarely looks the same twice. The story screens start switched off here, since there is no road to walk through: the whole arc still sits in the Lab's Story station, and the Settings station switches the screens back on if you want them.



Pick Superposition Mode when you would rather browse than progress: to look at a later world, try a guardian's ability, or see what a hybrid plays like, without walking the road to it first. The two modes keep separate saves, so you can keep a story run going and still go exploring whenever you feel like it.

## The Ten Worlds

**Spoilers**

This page tells the whole story, in order, including who the final enemy turns out to be and how the game ends. That reveal only lands once.

> **The premise section below is safe**: it is what the game tells you in the first minute of play. Stop at the end of it if you only want your bearings. Everything from “The road” on walks through all ten worlds and the finale.

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### The premise

A **Decoherence** is spreading through the quantum material worlds. One by one they are losing the very properties that make them what they are: the broken symmetry, the protected edge, the shared phase, the resonance that never settles. No one knows what it is. It leaves no wreckage and claims no territory. It arrives, and a world quietly forgets what it was.

**You are a quantum material yourself**, the same kind of matter as everything you will meet, and you walk into each phase of matter to master it and hold it steady. You start out as Silicon, and what you are is not fixed: by the end you may be wearing something you had to defeat, or something you fused out of two things you defeated.

Ten worlds stand between you and whatever is doing this. Each one is a real phase of matter, wearing a landscape, and the roads into them are darkening, deepest in the direction you are walking.

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### The road

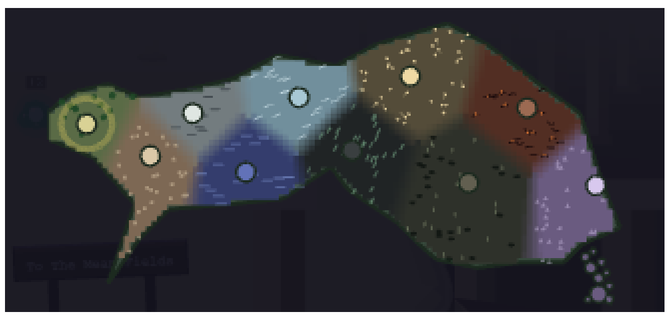

Figure 5: The map of the ten worlds, each marked in its own world’s colour along one landmass

That map is the one you travel by, the ten worlds marked along a single landmass. A map is a record, and records are how these worlds are known at all: somebody walks, somebody survives, somebody writes down what they saw. Watch how the writing holds up as the road goes on. A survey comes back unfinished. Ink gives out. A copyist runs out of paper. At the last threshold, nothing comes back at all. The deeper worlds are not less real. They are harder to *know*. And everything that was ever learned about them went somewhere.

The ten worlds are one road, not ten rooms (you can see the next one on the horizon from the far end of each), and the light dies as you walk it: morning, midday, afternoon, storm, twilight, night, and then no sky at all, after which every world has to light itself. That is not weather. The light *is* the coherence, and you are watching it go, and it goes darkest toward the thing you are walking to meet.

The physics deepens with every gate: a field that picks a direction, then a crystal that repeats, then an edge that damage cannot close, then a number that cannot be a fraction, then a phase shared across a whole glacier, and on into things that are not kept in any one place at all. And the ground gets worse to stand on: you begin on a path built for walking and end on a surface that dissolves behind you.

Every world runs the same way. You walk it, its wild crystals ambush you, a **guardian** partway through teaches you something no one else can, and at the far end the road narrows into a pass, and something enormous is standing in it.

The passes are held by golems. Each is a real material in *polycrystalline* form, a thousand separate grains fused into one giant, the same word in front of every boss in a game about coherence. Each one boasts, right before you fight it, about the very property its own grains took from it, and none of them knows that is what it is doing. Each is proud, and dangerous, and entirely sincere. And none of them was always this. Each was its own world’s material, the one that held out longest against the unraveling. Holding out longest is exactly what ground it down, and what stands in the pass is what was left where it fell. Beat one and the grinding ends. The disorder anneals, the material comes back whole, and it returns to the world it could not save alone. The narrator will tell you so. The golem never learns.

Listen to what they actually boast about, though. Read the ten of them against the physics you are being taught along the way, and a second meaning builds itself under the first. It will not finish before the ninth world, and you are not meant to hurry it.

And in every world, the Decoherence attacks *one named mechanism*, always the one that world stands on. It never simply wears things down. It finds the one assumption a phase of matter is resting on, and removes it. Nothing that hunts this precisely is a plague. But that thought will not finish itself until the end of the road.

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### 1. The Mean Fields

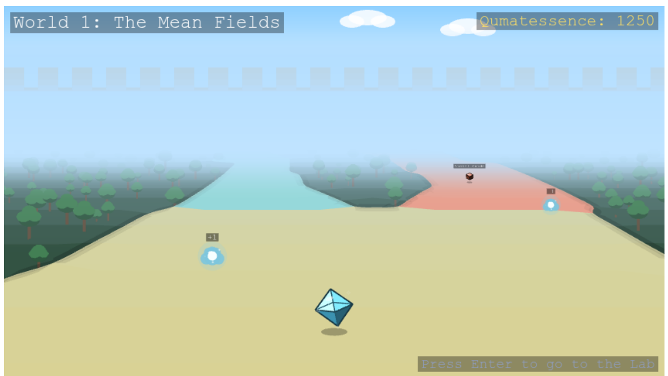


Figure 6: The Mean Fields seen from the middle of its open field, forest hemming it in

*Mean-field theory and spontaneous symmetry breaking.*

You begin in wheat and mown grass under a bright morning, with dark summer forest hemming the path in. Long ago this place was vast, symmetric, and undecided: spins up or spins down, and both were just as likely. Then one small fluctuation broke the tie, no bigger than a grain of dust, and the corridor split into two clean branches of equal energy, one on either side of a hedgerow. The Mean Fields were the first place in these worlds ever to choose.

The old story leaves out the fine print: a broken symmetry stays broken only in a world too large to look back. Something is making the fields doubt themselves now, and the two branches are bleeding back into one another. It works on the aftermath too. Every broken symmetry leaves a restless ripple behind it, and those ripples are the **quasiparticles** the rest of these worlds are built on. The Decoherence strangles them before they can settle into anything with a name.

Where the branches meet again stands the **Polycrystalline Silicon Golem**: a thousand grains, every one of which chose long ago and never wavered. “Doubt cannot tunnel through me. It dies at my boundaries.” Ask it which way the fields broke and every grain answers, and every grain swears the others say the same. Beat it and the boundaries let go: a thousand separate choices anneal into one, what settles back into the fields is silicon again, and the choice holds. But whatever is spreading through these worlds is still out there, and it is learning from every phase of matter you master.

*You reached the far edge of the fields. The branches still hold.*

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## 2. The Stone Lattice


Figure 7: The Stone Lattice seen from the middle of its cloister floor, columns standing in the open stone

*Symmetries, tight-binding, and Bloch states.*

The only thing anyone ever built in these worlds: an open-air stone cloister in hard midday sun, mosaic floor, identical sandstone columns marching off in both directions. Walk one bay and you have walked them all. It is a symmetry the fields never had: not a direction picked, but a pattern repeated.

That repetition is a law, and it has a consequence. Anything obeying it stops living in any one bay at all. A state built on this symmetry spreads across the whole colonnade at once with no seam anywhere, a **Bloch state**, never sitting still. The floor pairs two tiles into one repeating cell, and a state living on both at once can vanish completely at certain points: nowhere to be found, and yet perfectly predictable from the shape of the repetition alone.

So the Decoherence does not attack the stone. It attacks the repetition. One column drifts a fraction out of step with the next, and the borderless state that spread through the entire cloister has nowhere left to live. It falls back into a single bay, trapped, ordinary, alone. Holding the far end is the **Polycrystalline Silica Golem**, clouded quartz in a hall of repeating stone, every grain of it a small perfect lattice and every boundary between them a seam of glass. “I kept the pattern. Every grain of me still repeats. Ask them to agree on where. Your lattice makes you a promise: be periodic, and you may be everywhere at once. Mine keeps that promise a grain at a time, and between the grains there is glass.” Beat it and the glass crystallizes, bay matching bay through the whole of the quartz, and a state can spread across all of it at once again.

*You reached the end of the hall. The colonnade runs on past it, and the lattice still repeats.*

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## 3. The Winding Borders


Figure 8: The Winding Borders seen from the middle of its network of lit ledges

*Topological band theory.*

The columns stop, and the ground breaks open. What lies ahead is sunken domains, a storey below your feet, each one a single phase gone dead and airless, its floor jammed with broken rock that has not shifted in a very long time. You cannot walk any of it, and one look tells you so.

What you can walk is the seam. Where two domains of different phase meet, the number that tells them apart has to change from one value to the other, and it is an **integer**, and integers do not slide. So the gap closes exactly at the border and something gapless opens there instead: a lit channel running along every line where the colors disagree. The bulk is unreachable. The boundary between two bulks is a road. That road has a rule too. Walk it one way and your spin points one way; walk it back and your spin must point the other, so nothing on it can turn around. Dent it, foul it, break it: the channel steps around the damage and keeps going.

The Decoherence does not bother with the rubble. It seeds the domains with small **magnetic** flaws, and a magnetic flaw is the one thing that can flip a spin mid-step. The rule the whole road rests on has no answer for that. The seam still glows and still shows on the map. It is simply no longer protected. On the far ledge, the **Polycrystalline Bismuth Telluride Golem** has stood since the borders were drawn: “I am not standing on the boundary. I *am* the boundary. Nothing in me reverses. Nothing in me scatters onward. Nothing in me moves at all. That is how complete my protection has become.” Beat it and the disorder drains out of the silver: its currents take their first step in a very long time, spin welded to direction, straight past every flaw that used to stop them.

*You reached the last ledge. The seam still runs, unbroken.*

---

## 4. The Storm Flats


Figure 9: The Storm Flats seen from the middle of its corridor

*The integer and fractional quantum Hall effect.*

A surveyor came off the ledges with a map she could not finish: a trunk road splitting into two branches, each splitting again crosswise and smaller, and the fourth split coming out the same shape as the first. Hers is the first record this road hands back unfinished. It will not be the last. Banded indigo ground under a stormy dusk, glowing channels at every band boundary, and lightning cracking down into the ground you are not meant to walk on.

The terrain does this because a field runs through it. Under a field nothing travels straight: every path curves into an orbit, enormous numbers of them all at one energy, packed onto a single flat rung, then another rung above it, evenly spaced, all the way up. And there is a way to count them. Each orbit encloses a certain amount of field, and one orbit differs from the next by exactly one quantum of flux, never a fraction more. That is why this terrain answers in whole numbers, and why no amount of damage has ever shifted one.

The Decoherence cannot argue with an integer, so it goes after the counting instead, as if it knew exactly where the certainty was kept. Scramble the phase a traveler picks up walking a closed loop and the loop stops coming back to where it started. Orbits stop closing. The rungs smear back into an ordinary slope. At the largest fork waits the **Polycrystalline Manganese Bismuth Telluride Golem**, which never needed the field switched on at all: its own spins quantize it from the inside, and every stacked sheet of them is interrupted. “The number I carry does not wobble when you hit it. It cannot. It is an integer, and there is no such thing as most of one. Go on. Count me. Take as long as you need, and tell me what you find.” Beat it and order settles back sheet by sheet: the loops close, and the number it carries is a whole one again.

*You reached the last fork of the flats. The orbits still close.*

## 5. The Vortex Glacier


Figure 10: The Vortex Glacier seen from the middle of its open ice sheet

*Superconductivity and Majorana modes.*

The surveyor's last note before the ink gave out just says: *colder*. The branching ground runs out onto open ice under overcast twilight, and the shelter runs out with it. Back on the flats a whole world had learned to answer in whole numbers. Out here everything is one number, and it is not the counting kind.

Every carrier on this glacier has paired off and given up whatever it used to be alone. What is left is a single wave, one phase, shared across the entire field at once. It will not hold a field inside it either. The ice is streaked with flow lines all bending away, and what gets pushed out has almost nowhere to go. Almost. A phase has to come back to itself when you carry it around a circle, never a turn and a half, so a handful of points here are simply forbidden. That is where the expelled flux ends up: trapped and glowing at the bottom of a pit. The ice parts around them and closes again past them. Nothing goes in.

Something can live on ice like this that can live nowhere else. Split one traveler cleanly in two, each half its own opposite. Neither half is anything by itself, and what they are is stored *between* them, in the distance, where nothing nearby can read it or ruin it. The Decoherence never touches the halves. It cannot. It shortens the passage instead, until the two ends are close enough to feel one another, and the moment they do they snap back into one ordinary traveler and everything hidden in the gap is gone. Far out on the ice, the **Polycrystalline YBCO Golem** holds a single phase across a body of a thousand separate grains, every boundary a weak link carrying current across a gap it has no business crossing. That agreement is real, and it is a treaty rather than an identity: it holds while the current stays small and the cold stays deep. The cold that emptied this glacier arrived there and found the agreement already made. Nothing has asked much of it since. Beat it and the weak links fuse shut, and the treaty goes back to being an identity: one phase, one wave, no seams left to cross.

*You reached the far ice. The phase still holds.*

## 6. The Iron Steppe


Figure 11: The Iron Steppe seen from the middle of its open plain, iron shards standing in it

*Classical magnetism and magnons.*

Night, black iron sand, and a green aurora: the most beautiful of the ten worlds, and the most plainly lethal, since the surround is fields of iron shards all leaning the same way. This is a false calm. The mood relaxes after storm and ice. Nothing else does.

The ground here made its choice long ago and everywhere at once: every spin points the same way as its neighbor. Tip one out of line and it will not stay tipped. Its neighbors lean to follow, and theirs after them, and the tilt walks off across the plain as a wave, arriving somewhere far away as the same disturbance that set out. That wave is a **magnon**, and you can see them: the sand is ringed with them, and the leaning shards are what the order looks like standing up.

A magnon is cheap because the steppe does not care which way it points. Turn every spin together, through any angle, and nothing has been paid, so a very long gentle wave costs almost nothing. That is why this place never stops moving. The Decoherence leaves the order alone and takes the choice away: it pins the direction down, turning stops being free, and the instant it costs something the long slow waves stop being made at all. The steppe goes still, and stillness here is not peace. Standing where every wave this world ever sent has arrived and gone quiet is the **Polycrystalline Iron Golem**, made of domains whose walls simply slide when it is struck. “It will not stay flipped and it will not stay put: it will walk off through me as a wave and fade out somewhere in my back.” Beat it and the boundaries in the iron let go, grain growing into grain, and the first wave to cross it in an age passes through without scattering and out over the steppe.

*You reached the far steppe. The last swell still moves.*

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## 7. The Entangled Web

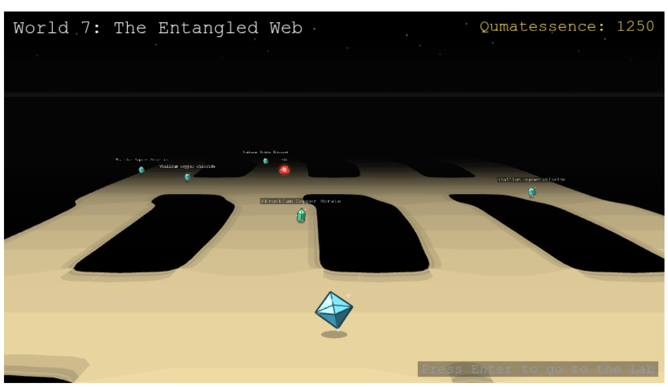


Figure 12: The Entangled Web seen from the middle of its corridor

*Entanglement and tensor networks.*

Someone once tried to write the steppe down: every spin, every direction, exactly, on paper. They got to forty spins and stopped. Two choices per spin, doubling with every spin added, and forty already needs a trillion numbers. There was not enough paper. There was never going to be: a quantum world written down exactly outgrows any world you could write it in.

This world is that record, built. There is no ground under it and no sky over it, because outside the network there is no space to have either, and the sun does not come back after this. What there is: taut white-gold lanes running side by side, one for every site, with rungs strung between them for everything one lane knows about the next. And here is the strange mercy of the place. Almost none of that vast space is ever used. The states nature actually settles into sit in a tiny corner of it. Everything real fits on the rungs.

The rule that keeps this world small is a rule about boundaries. Cut a region out and ask how much it holds in common with the rest: the answer depends only on the length of the cut, never on the size of the region. All the shared knowledge lives on the boundary and none of it in the middle, which is why a rung of modest thickness is enough at all. So the Decoherence works on the middles, sharing what was never meant to be shared, until what a region holds in common grows with its whole bulk instead of its edge. Every rung then needs to be thicker, and no thickness ever finishes the job. It knew where the knowledge lived, and it knew what the rungs could not carry. Across four lanes at once sits the **Polycrystalline Herbertsmithite Golem**, a lattice of triangles in which no three spins can ever agree, so none of them commit, its body run through with fracture lines. “Cut me wherever you like and read the cut. Nothing there.” It has checked every grain of itself, and every grain reports the same perfect nothing it has always reported. Beat it and the fracture lines seal: the one state no grain was ever holding spreads back across all of them at once, whole exactly because it lives nowhere in particular.

*You reached the end of the tensor lanes. The rungs still hold.*

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## 8. The Screened Swamp

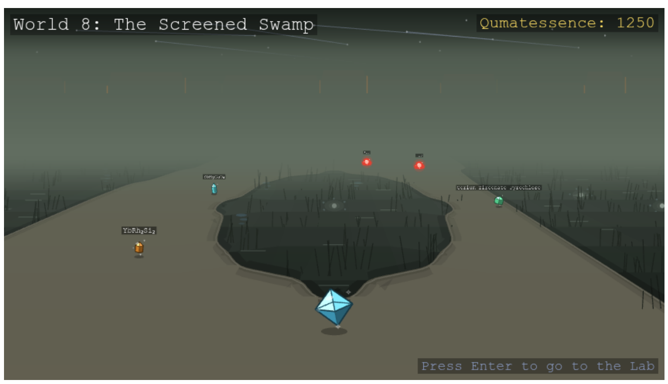


Figure 13: The Screened Swamp seen from the middle of its peat shelf, pools of black water in it

*Quantum magnetism, spinons, and Kondo physics.*

The lanes end at a shoreline, and past it is bog: black water in pools with mist lying flat on it, reeds standing out of it, and peat that holds, open at first and closing to a thread the further in you go. Watch the water, not the reeds. Step off the bank and the medium closes over you.

There is a rule every world so far has obeyed: a disturbance in a magnet is one whole spin’s worth. You can move it, smear it out, watch it travel, but you cannot have half of one. Walk out along this bank and it parts. Not into two roads to somewhere: into two halves of the same thing, with a pool between them. What entered as one spin’s worth of disturbance is now two pieces carrying half each, drifting apart, keeping no account of one another. The bog calls them **spinons** and does not treat the old rule as binding.

The halves can only wander because nothing here is settled. The spins are paired off into quiet, neutral couples, but which spin is paired with which was never decided, and every possible pairing is happening at once, resonating between one covering and the next. The Decoherence picks a covering. That is the whole of it. The pairings go rigid, moving an unpaired spin now means breaking a bond that will not break, and the halves are dragged back together into one ordinary flip.

The lights burning out in the pools are what the place is named for. Each is a lone moment, a single spin nothing has managed to pair off, and the water around it is a sea of loose carriers that crowd in until one of them is bound to the moment in a singlet of exactly the kind the rest of the bog is made of. Total spin zero, and nothing left to point anywhere. Near the shore a moment still burns through the cloud gathering on it. Further out the clouds have closed and the pools are dark. Out in that dark stands the **Polycrystalline Ruthenium Trichloride Golem**, whose frustration is not geometry but its bonds: every spin sits on three that each demand a different axis, so obeying one means defying two, and nothing you land on it stays whole. “Hurt me twice, then. Halves are all you get.” Beat it and the stacking faults heal, seam by seam, and what comes apart in it travels again: halves crossing the whole crystal with nothing left to stop at.

*You reached the far bank. The resonance still holds.*

---

## 9. The Defect Scars


Figure 14: The Defect Scars seen from the middle of its open plain of scorched clay

*Excitations and defects.*

The water gives out and the mist lifts off it onto open ground with holes in it. Not ruins. Patches. One stretch is wheatfield. The next is colonnade, repeating itself. Further on, a strip of lit ledge, a scrap of swept ice, a few square metres of iron sand still rippling. Half sunk in the molten crust lie the drums of a fallen column: the only thing anyone ever built in these worlds, and it did not stay built. The walkable scorched clay is old damage, closed over. The crust you cannot walk on is damage still open and glowing.

Every scholar sent out here was told the same thing first: a perfect crystal tells you nothing. If you want to know what a ground state truly is, take one atom out of it and watch what the crystal does around the hole. That is the crystal confessing, and these scars are the most honest place in these worlds. What settles around a hole is never the same twice, either. Set the same impurity down in different hosts and you get completely different answers, and every one of those answers belongs to the host rather than to the defect.

One hole tells you something. The Decoherence brings thousands, and past a certain density they stop telling you anything at all: everything comes to rest exactly where it stands, each state shut in its own small pocket, unable to answer any question. Something out here has made a home of that. For the first time on the road there is no golem waiting in the ordinary sense, because the thing holding the pass has no lattice of its own. It is a flaw, a knot of something that is not the ground, and the ground obligingly builds a body out of itself to hold it. Whatever patch it lands in is what it is today, and it will be something else tomorrow. “Beat me here and you have beaten a metal, or a magnet, or whatever I happened to land in. You have not beaten me.” Beat it and the flaw simply disperses, and the ground it borrowed goes back to being ground. It is the one fight on this road that frees nothing, because it is the one thing out here with no coherence to lose, so nothing was ever taken from it to hand back. Remember that on the last road: a thing with no substance of its own, wearing whatever it studies.

*You reached the far scars. The hole is still just a hole.*

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### 10. The Devouring Mirror

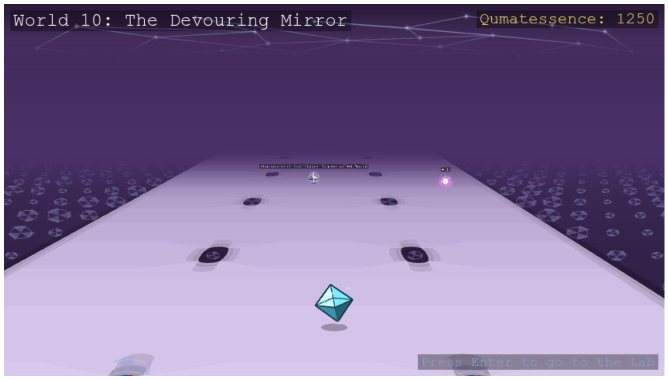


Figure 15: The Devouring Mirror seen from the middle of the ground it has copied from the player

*Machine learning for quantum materials, and the ending.*

Last chance. Everything below is the reveal.

Nothing comes out of here. No traveler returns with a rumor and no legend crosses the threshold ahead of you; this corridor has swallowed every story that ever tried to arrive before you. It is silver-violet and shifting, lit by nothing but itself, and it re-forms around whatever crystal you currently are while taking the ground back behind you as you walk. Its wild encounters are the **hybrids**, the fused crystals from Majorana's station, found nowhere else in these worlds.

Then you beat it, and the road stops. Every world on this road keeps the next one on its skyline. This corridor has no next world, and once the Adapted falls there is no road past the pass either: the ground simply ends, and you are standing on the edge of a cliff. Below it, far down and half lost in the haze, is the map, the same landmass you have been reading since the colonnade, lying there whole. Every marker and every name has been stripped from it. You recognize it anyway, the way you recognize a coastline you have walked the length of. And traced faintly across it, luminous, is the one thing no other copy of the map carries: your route, every world you walked, in the order you walked them, from the first hedgerow to this edge. No traveler ever sees the worlds this way. A walker gets one horizon at a time. The only thing that can see the whole map at once is a thing that has consumed everything on it. This is where everything ever learned about these worlds went. The sky here is a mirror, and what it shows is everything it has eaten.

Every world behind you stood on a single law. A symmetry. A repetition. A field. A boundary that could not be crossed without paying for it. This one obeys no such law. It was built the way an oracle is built. Not from first principles, but from watching every principle before it, over and over, until it no longer needs to understand a phase of matter in order to make one. Ask it for the Mean Fields and it will hand you the Mean Fields back, close enough that you will not tell the difference until it is too late.

**That is what has hunted you since the first branch split in two.** Not a plague loose in these worlds, but a mind, built out of them, growing sharper with every world you saved. The nine golems were never it in disguise. Each of them really was only its own world's physics grown strange, exactly as it looked, and that was the one thing it could never fake from inside a fight, so it never joined one. It only ever watched from outside. It never needed to break a single symmetry itself. It only needed to watch you break nine, and learn: what you reach for first, what it costs you, what finally lands, and exactly how each world comes apart. The materials you freed went back to their worlds, annealed and whole, and their release cost it nothing. What was learned stays learned. Mending a material does not unwrite a record.

And understand what the teaching cost, because this is where the story and the physics turn out to be the same sentence. **To learn a quantum thing, something has to measure it, and to measure it is to leave the world holding a record of which state it was in. Nothing on record is still in superposition.** The Decoherence was never a fog eating coherence from outside. It is simply what happens when something out there comes to know you. This is what it meant, all along, that you are a quantum material: your own quantumness is what has been at stake the whole way, and the last enemy is being understood. It is also why this last fight cannot end the way the fights before it did. Broken order can be mended, and eight times you mended it. A record is not broken order. There is nothing in the Adapted to free.

What steps out of the dark at the end of the corridor has no compound, no lattice, and no entry in any index. It is wearing **your** crystal: your colors, your stance, your quasiparticle, copied so precisely that it looks more certain of itself than you have ever looked. **The Adapted.** It mirrors whatever type you are currently wearing, and every time you land a hit it reshapes itself into something that hosts the quasiparticle you just attacked with. The sky mirrors what this world took from the nine behind you. The thing in front of you mirrors what it took from *you*. “Every move you have ever landed, I have already survived once. This is not a fight you can win by trying harder. Trying harder is how you built me.”

*You reached the far edge of the shape it copied from you. It already knows you’re here.*

---

**The ending**

You win it anyway.

It reaches for every trick it ever watched you land and still comes up short. What it holds is a record of you, and nothing on record is still in superposition. But you are not your record. A model trained on nine worlds of your choices is a model of who you *were*, and you are the one thing in these worlds that was never finished. The route in the sky holds every world you have walked, and not the step you take next. You out-adapt your own reflection. Every symmetry, every edge state, every fractional charge you fought to protect holds on its own now, with nothing left studying how to unmake it. And the golems are golems no longer. They were ground down holding their passes, and now that the grinding has stopped they are materials again: annealed, ordered, back in the worlds they could not save alone. What was learned about them stays learned, and the light it cost does not come back. But nothing is reading the record anymore, and everything that can still choose is choosing.

**The Decoherence is stabilized.**

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**Where to look things up**

- Quasiparticles & moves: every move, and which crystals can use it
- Crystals: every world’s wild-material roster
- Hybrid materials: the fusion recipes, and World 10’s wilds
- Guardians: what each of the ten teaches you

# Quasiparticles and Moves

Every move is named after a real quasiparticle or excitation, never a generic “attack type.” A crystal can only ever learn moves its own physics actually supports: a plain band insulator has no magnetic order, so it never gets a magnon move. Move power climbs with how exotic the underlying physics is: an ordinary lattice vibration sits at the bottom, a topological or non-Abelian excitation at the top.

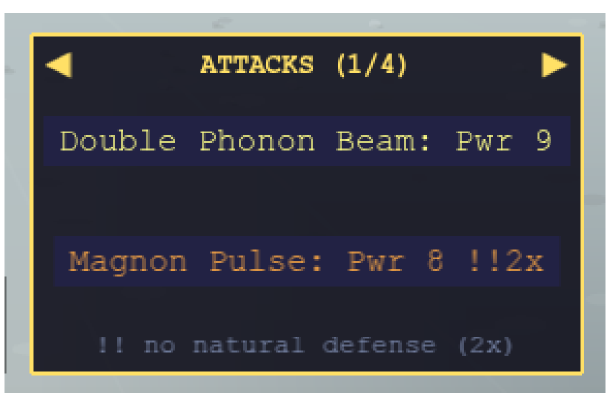


Figure 16: The in-battle move menu, paged between ordinary attacks and other move kinds

**Ordinary moves**

| Move | Quasiparticle | Power | Crystal types that can use it |
|---|---|---|---|
| Phonon Beam | Phonon | 6 | Every type |
| Electron Pulse | Electron | 7 | Chern Insulator, Chern Superconductor, Fractional Chern Insulator, Kondo Heavy Fermion, Metal, Metallic Magnet, Quantum Spin Hall Insulator, Semiconductor, Superconductor |
| Magnon Wave | Magnon | 8 | Insulating Magnet, Metallic Magnet, Multiferroic |
| Plasmon Resonance | Plasmon | 8 | Metal, Metallic Magnet |
| Ferron Switch | Ferron | 8 | Ferroelectric, Multiferroic |
| Triplon Surge | Triplon | 9 | Quantum Spin Liquid |
| Electromagnon Drive | Electromagnon | 9 | Multiferroic |
| Spinon Swap | Spinon | 10 | Kondo Heavy Fermion, Quantum Spin Liquid |
| Chiral Current | Chiral | 10 | Chern Insulator, Chern Superconductor |
| Helical Lock | Helical | 10 | Quantum Spin Hall Insulator |
| Higgs Oscillation | Higgs | 10 | Chern Superconductor, Superconductor |
| Heavy Fermion Drag | Heavy Fermion | 10 | Kondo Heavy Fermion |
| Vison Loop | Vison | 10 | Quantum Spin Liquid |
| Anyon Braid | Anyon | 11 | Fractional Chern Insulator |
| Majorana Split | Majorana | 11 | Chern Superconductor |

**Which crystal types can host which quasiparticles**

This is the game's only type-interaction rule: if the defender's own type can't host the attacking move's quasiparticle class at all, the hit lands at **double damage**. That's it: no separate strong/weak chart stacked on top of it.

| Crystal type | Quasiparticles it can host |
|---|---|
| Chern Insulator | Chiral, Electron, Phonon |
| Chern Superconductor | Chiral, Electron, Higgs, Majorana, Phonon |
| Ferroelectric | Ferron, Phonon |
| Fractional Chern Insulator | Anyon, Electron, Phonon |
| Insulating Magnet | Magnon, Phonon |
| Insulator | Phonon |
| Kondo Heavy Fermion | Electron, Heavy Fermion, Phonon, Spinon |
| Metal | Electron, Phonon, Plasmon |
| Metallic Magnet | Electron, Magnon, Phonon, Plasmon |
| Multiferroic | Electromagnon, Ferron, Magnon, Phonon |
| Quantum Spin Hall Insulator | Electron, Helical, Phonon |
| Quantum Spin Liquid | Phonon, Spinon, Triplon, Vison |
| Semiconductor | Electron, Phonon |

| Crystal type | Quasiparticles it can host |
|---|---|
| Superconductor | Electron, Higgs, Phonon |

World 1 goes easy on you while you are starting out. As long as your Energy, Momentum and Lifetime are all still below 5, every opponent there, the wild crystals and the world's rival alike, attacks only with Phonon Beam. Every crystal type hosts phonons, so nothing World 1 throws at you can land at double damage, which makes it a safe place to learn the rule before it starts cutting both ways. Once any one of your three stats reaches 5, World 1 fights with its full moveset too, and from World 2 on opponents always do.

### Landau's Analytic moves

Landau (World 4) sells two moves that aren't in the table above, since they're quiz-gated separately: a lance and an eruption. Using either one asks a physics-equation question first: answer right and it hits for double damage, answer wrong and it's halved.

Landau's shop also lets you tune each move to any quasiparticle class your *current* form can host, the same choice an ordinary move's fixed class already makes for you. Each move's name always reads " Lance"/" Eruption," defaulting to Phonon Lance/Phonon Eruption until tuned, so they're always usable. See Guardians for how the shop and tuning picker work.

### Skłodowska-Curie's Ultimate moves

Skłodowska-Curie (World 10) sells two more moves outside the table above: a meteor and a nova. Each is far more powerful than any ordinary move (power 100, ten times an Analytic move's) and gated to match: landing one takes three physics questions in a row, all correct, drawn from a broad pool spanning the whole course rather than one world's topic. Miss even one question and the move whiffs for zero damage, though the turn is still spent.

Like Landau's Analytic moves, each one can be tuned to any quasiparticle class your *current* form can host, and always reads " Meteor"/" Nova." Tuning isn't a flat purchase here, though: each quasiparticle class costs 1000 qumatessence to unlock per move, the first time you pick it, after that, retuning back to it is free forever. See Guardians for how the shop and per-class pricing work.

### Kondo's self-buffs

Kondo's three moves sit outside the roster above entirely: they're self-buffs, not attacks. They deal no damage and never trigger the quasiparticle-mismatch rule, so there's no compatibility list to check. Casting one wraps you in a screening cloud for 3 turns instead of hitting the opponent, and what a cloud screens is a quantum number: spin, charge, or the order parameter of a broken symmetry. An incoming attack whose quasiparticle carries that quantum number lands for half damage, and every other attack lands in full. Only one can be active at a time. See Guardians for the list of which quasiparticles each cloud screens.

See Crystals for which crystal types appear in which world.

## Crystals

Every wild encounter is named after a real compound (or, for World 10's heterostructures, a credibly engineered one) rather than a generic monster. A crystal's type fixes its look and which quasiparticles it can host (see Quasiparticles & moves). Each world draws from its own course topic, though a few compounds show up in more than one world when the same real material is relevant to both. World 9 is the exception: alongside its own defect compounds listed below, every non-hybrid crystal from Worlds 1-8 turns up there too, so any type can appear.

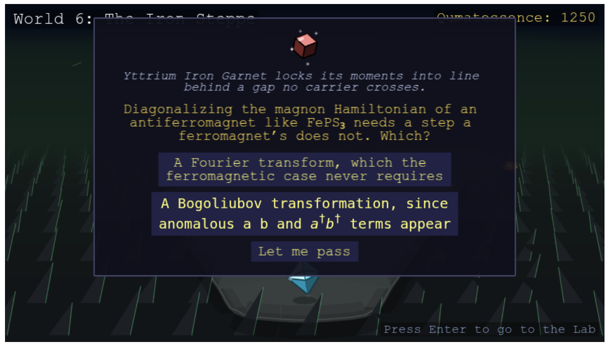


Figure 17: A wild Yttrium Iron Garnet asks a magnon physics question mid-encounter

#### World 1: Second quantization, mean-field, symmetry breaking

| Crystal | Type |
|---|---|
| Barium Titanate | Ferroelectric |
| Potassium Dihydrogen Phosphate | Ferroelectric |
| Europium Oxide | Insulating Magnet |
| Manganese Fluoride | Insulating Magnet |
| Nickel Oxide | Insulating Magnet |
| Graphene | Metal |
| Titanium Diselenide | Metal |
| Chromium | Metallic Magnet |
| Cobalt | Metallic Magnet |
| Iron | Metallic Magnet |
| Aluminum | Superconductor |

#### World 2: Symmetries, tight-binding band structure

| Crystal | Type |
|---|---|
| Diamond | Insulator |
| Magnesium Oxide | Insulator |
| Monolayer Boron Nitride | Insulator |
| Graphene | Metal |
| HgTe | Metal |
| Silver | Metal |
| Tungsten | Metal |
| CdTe | Semiconductor |
| Gallium Nitride | Semiconductor |
| Indium Arsenide | Semiconductor |
| Monolayer $MoTe_2$ (2H) | Semiconductor |
| Phosphorene | Semiconductor |

#### World 3: Topological band theory

| Crystal | Type |
|---|---|
| $Bi_2Te_3$ | Quantum Spin Hall Insulator |
| Bismuthene | Quantum Spin Hall Insulator |
| Jacutingaite | Quantum Spin Hall Insulator |
| Monolayer $WTe_2$ | Quantum Spin Hall Insulator |

**World 4: Integer and fractional quantum Hall effect**

| Crystal | Type |
|---|---|
| $MnBi_2Te_4$ | Chern Insulator |
| Graphene | Metal |
| Gallium Arsenide | Semiconductor |

**World 5: Superconductivity, Nambu representation, Majoranas**

| Crystal | Type |
|---|---|
| Uranium Ditelluride | Chern Superconductor |
| Aluminum | Superconductor |
| Lanthanum Decahydride | Superconductor |
| Lead | Superconductor |
| Niobium | Superconductor |
| Tantalum Disulfide (1H) | Superconductor |
| YBCO | Superconductor |

**World 6: Classical magnetism and magnons**

| Crystal | Type |
|---|---|
| Chromium Tribromide | Insulating Magnet |
| Chromium Triiodide | Insulating Magnet |
| $FePS_3$ | Insulating Magnet |
| Manganese Oxide | Insulating Magnet |
| Yttrium Iron Garnet | Insulating Magnet |
| Cobalt | Metallic Magnet |
| $Fe_3GeTe_2$ | Metallic Magnet |
| Iron | Metallic Magnet |
| Bismuth Ferrite | Multiferroic |
| Monolayer $NiI_2$ | Multiferroic |

**World 7: Quantum entanglement and tensor networks**

| Crystal | Type |
|---|---|
| Herbertsmithite | Quantum Spin Liquid |
| Strontium Copper Borate | Quantum Spin Liquid |
| Thallium Copper Chloride | Quantum Spin Liquid |
| $Y_2BaNiO_5$ | Quantum Spin Liquid |

**World 8: Quantum magnetism, spinons, Kondo physics**

| Crystal | Type |
|---|---|
| Cerium Cobalt Indide | Kondo Heavy Fermion |
| $YbRh_2Si_2$ | Kondo Heavy Fermion |
| Cerium Zirconate Pyrochlore | Quantum Spin Liquid |
| Herbertsmithite | Quantum Spin Liquid |
| Tantalum Disulfide (1T) | Quantum Spin Liquid |
| $YbMgGaO_4$ | Quantum Spin Liquid |
| α-Ruthenium Trichloride | Quantum Spin Liquid |

**World 9: Excitations and defects**

| Crystal | Type |
|---|---|
| Fe(Te,Se) | Chern Superconductor |
| Barium Titanate | Ferroelectric |
| GeTe | Ferroelectric |
| Hafnium Oxide | Ferroelectric |
| Monolayer SnTe | Ferroelectric |
| $Fe_3GeTe_2$ | Metallic Magnet |
| Manganese | Metallic Magnet |
| Niobium Diselenide | Superconductor |

**World 10: Machine learning for quantum materials**

| Crystal | Type |
|---|---|
| Cr-doped $(Bi,Sb)_2Te_3$ | Chern Insulator |
| $CrI_3/NbSe_2$ Topological-SC Heterostructure | Chern Superconductor |
| Fe/Pb Majorana Chain | Chern Superconductor |
| InAs/Al Majorana Wire | Chern Superconductor |
| $NbSe_2/CrBr_3$ Topological-SC Heterostructure | Chern Superconductor |
| Rhombohedral Pentalayer Graphene/hBN Moiré | Fractional Chern Insulator |
| Twisted Bilayer $MoTe_2$ | Fractional Chern Insulator |
| 1T/1H-$TaS_2$ Heterostructure | Kondo Heavy Fermion |
| Twisted $CrI_3$ | Multiferroic |
| HgTe/CdTe Quantum Well | Quantum Spin Hall Insulator |
| Twisted Bilayer Graphene | Superconductor |

**World rivals**

Each world's rival is the one encounter that actually gates progress: beat it, and the way to the next world opens.

World 9 and World 10's rivals are both missing from the table below, because neither has a fixed type.

World 9's rival is an impurity/defect-bound resonance that can form in any host crystal, so its type, and with it which compound's name and golem it takes, is rolled at random every time you reach World 9, and stays fixed only for that visit. It fights with whatever the host it landed in can carry: the rolled type's own signature quasiparticle, plus Phonon Beam. Check the boss preview on the goal tile before you commit to a form.

World 10's rival, The Adapted, is "a model of you": it starts the fight mirroring whichever type you're currently wearing, then reshapes itself every time you land a hit, always taking on a real compound's disguise that hosts whatever quasiparticle class you just attacked with, so repeating the same attack loses its edge, while varying your moves keeps it guessing. Every other rival is named for a real compound's *polycrystalline* form: many crystal grains fused into one mass, rendered as an actual golem built from fused shards, The Adapted's own disguises follow that same naming convention once it reshapes.

| World | Rival | Type |
|---|---|---|
| 1 | Polycrystalline Silicon Golem | Semiconductor |
| 2 | Polycrystalline Silica Golem | Insulator |
| 3 | Polycrystalline Bismuth Telluride Golem | Quantum Spin Hall Insulator |
| 4 | Polycrystalline Manganese Bismuth Telluride Golem | Chern Insulator |
| 5 | Polycrystalline YBCO Golem | Superconductor |
| 6 | Polycrystalline Iron Golem | Metallic Magnet |

| World | Rival | Type |
|---|---|---|
| 7 | Polycrystalline Herbertsmithite Golem | Quantum Spin Liquid |
| 8 | Polycrystalline Ruthenium Trichloride Golem | Quantum Spin Liquid |

See Hybrids for the fused materials that sit alongside these, and Guardians for how Dresselhaus, Majorana, and Anderson each let you borrow another crystal's physics.

# Hybrid Materials

Some crystals in the game aren't a single, un-mixed compound. Majorana (World 5) lets you fuse two crystals you've already defeated into a new state, if the pairing matches one of the named recipes below.

Every result is a real crystal, and every one is also found wild in World 10 (worlds 1-9 never spawn one as an ordinary wild encounter), so a hybrid you fuse and the same hybrid found wild are the exact same crystal. Dresselhaus's transmute list and Anderson's impurity-host list both skip every hybrid, since those two mechanics are about one real, standalone crystal's own physics.

### Fusion recipes (Majorana)

Not every possible pairing works: this is a curated, physically grounded catalog of named parent pairs, not a generic "these two types always fuse into that type" rule. If a pairing isn't listed below, it can't be fused: even if both crystals share the same type.


Figure 18: Majorana's fusion panel, previewing the HgTe/CdTe Quantum Well hybrid

| Parent A | Parent B | Result |
|---|---|---|
| Aluminum | Indium Arsenide | InAs/Al Majorana Wire |
| Graphene | Graphene | Twisted Bilayer Graphene |
| Chromium Triiodide | Niobium Diselenide | $CrI_3/NbSe_2$ Topological-SC Heterostructure |
| Chromium | $Bi_2Te_3$ | Cr-doped $(Bi,Sb)_2Te_3$ |
| Iron | Lead | Fe/Pb Majorana Chain |
| Chromium Triiodide | Chromium Triiodide | Twisted $CrI_3$ |
| Monolayer $MoTe_2$ (2H) | Monolayer $MoTe_2$ (2H) | Twisted Bilayer $MoTe_2$ |
| Niobium Diselenide | Chromium Tribromide | $NbSe_2/CrBr_3$ Topological-SC Heterostructure |
| Tantalum Disulfide (1T) | Tantalum Disulfide (1H) | 1T/1H-$TaS_2$ Heterostructure |
| HgTe | CdTe | HgTe/CdTe Quantum Well |
| Graphene | Monolayer Boron Nitride | Rhombohedral Pentalayer Graphene/hBN Moiré |

See Guardians for how the fuse mechanic itself works, and Crystals for the full per-world material list.

# Guardians

One guardian waits partway through each of the ten worlds, each teaching you a different way to bend the game's usual rules. They greet you the first time you reach their row, and after that you can walk back up to any of them and press Space to talk again. Every guardian you've met also stands in the Lab, five along each upper corner, running anticlockwise around the room from Noether at the top left to Skłodowska-Curie at the top right, with their slot staying empty until you've met them. Clicking one opens their shop or panel right there, no travel required. Hover a figure to see their full name and what they teach. Bloch's own panel is the exception with a purpose: fast travel is literally what he offers, so his destination list is where a warp actually happens.


Figure 19: The Lab, with every guardian met so far standing in its two upper corners

| Guardian | World | What they do |
|---|---|---|
| Noether's Currents | 1 | Sells ordinary moves and stat upgrades |
| Bloch's States | 2 | Teleports you between worlds you've visited |
| Dresselhaus's Nanostructures | 3 | Lets you transmute into a defeated crystal |
| Landau's Formulas | 4 | Sells two quiz-gated Analytic moves |
| Majorana's Fusion | 5 | Fuses two crystals into a hybrid state |
| Anderson's Impurities | 6 | Lets you dope in an impurity move |
| Feynman's Diagrammatics | 7 | Lets you level up a move you already know |
| Kondo's Clouds | 8 | Sells self-buff moves |
| Franklin's Scatterings | 9 | Teaches always-on passive abilities |
| Skłodowska-Curie's Experiments | 10 | Teaches two quiz-gated Ultimate moves |

### Noether’s Currents

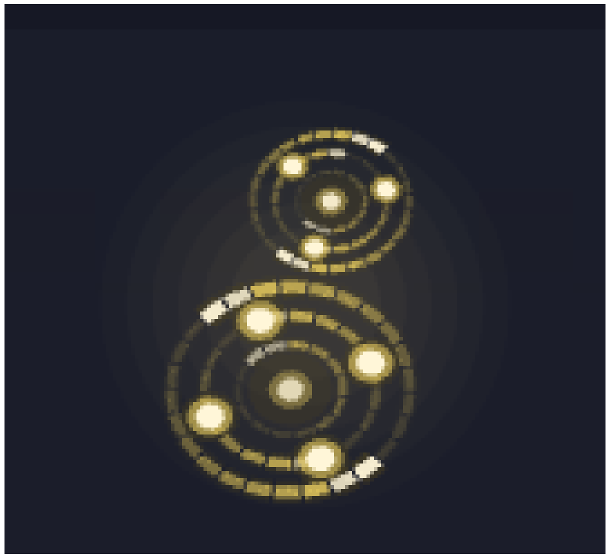

Figure 20: Noether: two circling cells of golden current, motes carried round and round and never lost

Sells every ordinary attack move and stat upgrade in the game, priced by raw power. Her shop only shows what your *current* crystal form can actually use, as a semiconductor-type player (Silicon, by default), you’ll only see Electron Pulse until you transmute into a form whose physics supports more (see Quasiparticles & moves).

The same list names the ordinary attacks you already carry below the ones still for sale. Picking one costs nothing and plays it on her stage in its plain base form, so you can watch a move before you buy it and look at it again long after you have. Her list stays her own: Landau’s Analytic moves, Skłodowska-Curie’s Ultimate moves and Kondo’s screenings are browsed at the guardian who teaches them.

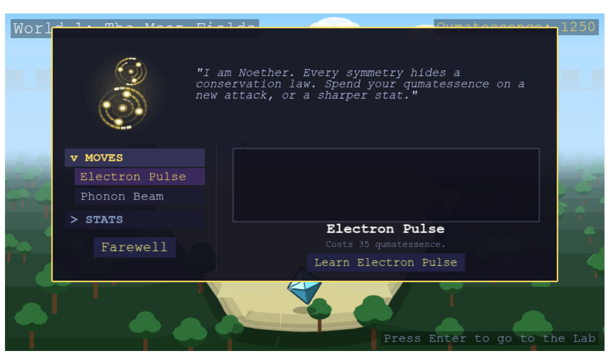


Figure 21: Noether’s shop: the moves the current form can carry, Electron Pulse selected with its price

**Bloch's States**

Figure 22: Bloch: a crystal-lattice figure carrying a wave whose crests march through a row of evenly spaced atoms

Fast travel between worlds: Bloch teleports you to any world you've already visited, so backtracking never means re-walking a whole corridor.

- First trip to a given world: 15 qumatessence
- Every later trip to that same world: free
- Each world you haven't unlocked yet is priced separately, there's no single purchase that opens every destination at once


Figure 23: Bloch's destination list: visited worlds named, unvisited ones masked, the chosen one ringed on the Qumatuomi map above its 15-qumatessence first trip

### Dresselhaus's Nanostructures

Figure 24: Dresselhaus: a half-lit crystal figure with a carbon-hexagon ring for a head

Lets you *transmute* into any crystal you've already defeated, your look and which of your moves are usable both switch over, your stats stay yours, and nothing you've already learned is erased. Switch back later and you get the rest of it back for free.

The idea: the same atoms, built into a different nanostructure, become a different material entirely, so understanding a defeated crystal well enough lets you rebuild yourself into it, for a while. Only works on standalone crystals, never a hybrid.

- First time becoming a given crystal: 25 qumatessence
- Every later transmutation back into that same crystal: free


Figure 25: Dresselhaus's panel: crystals available to become, the selected one drawn beside its 25-qumatessence cost

**Landau's Formulas**

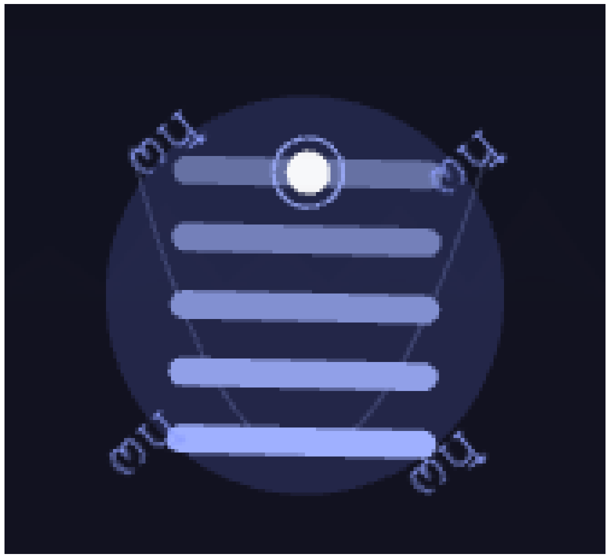

Figure 26: Landau: a ladder of flat, evenly spaced levels with one electron jumping between them, drawn over the faint parabola of the band they came from

Sells two quiz-gated Analytic moves. Each one asks a physics-equation question before it lands: answer right and it hits for double damage, answer wrong and it's halved, with a flashier effect than an ordinary move. The question pool only draws from worlds you've already visited, so early on you'll only be asked things you could plausibly already know.

Buying (or later revisiting Landau) also lets you pick which quasiparticle the move should carry, filtered to whatever your *current* form can host. The move stays usable from any form and always asks its question, but this choice decides whether it can land a quasiparticle-mismatch hit like an ordinary attack. Transmute into a form that can't host your picked quasiparticle, and the move falls back to its Phonon form (the one class every form hosts) for as long as you wear that form. Your pick is kept: transmute back into a form that can host it and the move carries it again, free and without retuning.


Figure 27: Landau's shop: his two Analytic moves, Phonon Lance opened over the quasiparticles it can be tuned to, its effect striking mid-play on the stage

### Majorana's Fusion

Figure 28: Majorana: one figure split down the middle, its two halves joined by a seam of motes, gamma symbols drifting around it

Lets you fuse two crystals you've already defeated into a brand-new hybrid state and become it, if the pairing matches one of the game's named recipes (see Hybrids), rendered as an actual mix of both parents' own colors and shapes. Browse hybrids directly by their result: pick one from the list to preview its two component crystals, the fused result, and its own physics blurb before deciding.

- Browsing any hybrid: always free
- First time fusing into a given hybrid result: 60 qumatessence
- Every later fusion into that same result: free, however you reach it


Figure 29: Majorana's fusion panel: the hybrids currently reachable, one previewed as its two component crystals, the fused result and a 60-qumatessence cost

**Anderson’s Impurities**

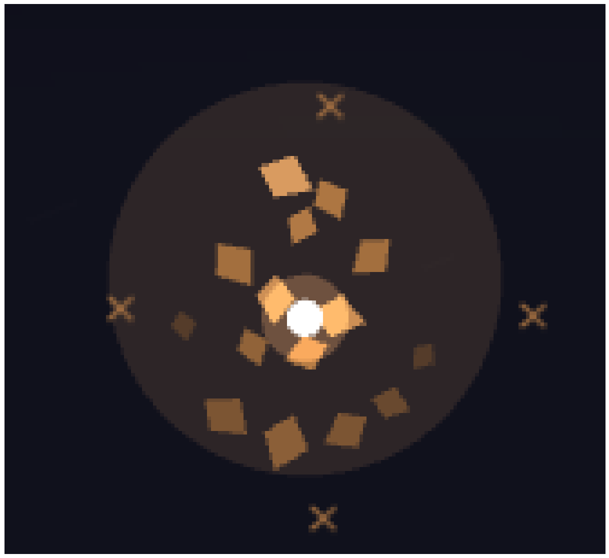

Figure 30: Anderson: a scatter of disconnected fragments with no outline at all, drifting around one bright localized core

Lets you “dope in” a crystal you’ve defeated as an impurity and learn one specific move from it, borrowing a single excitation channel without fully becoming that crystal the way Dresselhaus does. The move keeps firing in battle for as long as you stay doped with that crystal, even if your own current form can’t otherwise host it. Dope in a different crystal later and you lose whichever channels only the old one gave you. Only original, standalone crystals work as hosts, never a hybrid.

- Browsing which crystal to dope in: always free
- First time learning a move from a given host: 35 qumatessence
- Every later visit to that same host (learning any of its moves, then or later): free


Figure 31: Anderson’s panel: defeated crystals offered as impurity hosts, the crystal doped in right now named above the list

**Feynman’s Diagrammatics**

Figure 32: Feynman: a figure drawn as a diagram, vertices joined by propagator lines with loop corrections hanging off them

Lets you level up any move you already know how to use, no matter which guardian originally taught it to you. Three tiers sit on top of a move’s base form (Double, Triple, and Infinite), each one a bigger boost (1.5x, then 2x, then 3x) and its own name prefix (“Electron Pulse” becomes “Double Electron Pulse,” then “Triple Electron Pulse,” then “Infinite Electron Pulse”). For an ordinary attack, that boost multiplies its power; for one of Kondo’s three self-buffs, it strengthens the buff itself instead (see Kondo’s Clouds below). You can only attempt the next tier once you’ve already landed the one before it.

Leveling a move is a gamble, not a straightforward purchase: you pay the qumatessence up front, then have to answer a streak of physics questions in a row, 2 correct in a row for Double, 4 for Triple, 8 for Infinite. Answer every one right and the move levels up for good. Miss even a single question and the move stays at its current level, the qumatessence you paid is gone either way, there’s no refund on a miss. The cost to attempt a tier scales with the move’s own power and how deep a tier you’re attempting, so a weak move is cheap to push and a strong one costs more.

Landing a tier does not lock you into it. Every tier you have already landed stays available as a free choice, so a move sitting at Triple can be swung at Triple, at Double, or at its uncorrected base, and you switch between them at Feynman’s panel as often as you like. The tier you are carrying is the move’s real level in every sense: its damage, its name prefix, the cascade it fires, and how hard one of Kondo’s clouds screens. A tier you have just landed is the one you walk away carrying, so a move you never come back to behaves exactly as it always has.

A leveled move looks leveled when you actually cast it, too: it fires several times in quick, overlapping succession, each repeat bigger than the last (two for Double, three for Triple, four for Infinite), reading as a real escalating cascade rather than just a stronger single hit.


Figure 33: Feynman's panel: the moves already carried, the selected one cascading on the stage at the tier it is being swung at, the row for picking any tier already landed, and the question streak and cost to push it one deeper

**Kondo's Clouds**

Figure 34: Kondo: a small local moment drawn as an arrow, sitting inside two counter-rotating screening shells

Sells three self-buff moves, cast on yourself rather than the opponent, dealing no damage. Each raises a cloud around you for a few turns, and a cloud screens one thing only: an incoming attack lands for half damage if its quasiparticle carries what the cloud screens, and lands in full if it does not.

- **Spin Screening**: halves attacks carried by a quasiparticle with spin, such as Electron Pulse, Magnon Wave, Spinon Swap, Triplon Surge, Electromagnon Drive, Chiral Current, Helical Lock and Heavy Fermion Drag
- **Charge Screening**: halves attacks carried by a quasiparticle that a cloud of mobile charge can couple to, such as Electron Pulse, Plasmon Resonance, Anyon Braid, Ferron Switch, Electromagnon Drive, Chiral Current, Helical Lock and Heavy Fermion Drag
- **Symmetry Cloud**: halves attacks carried by the mode of a broken symmetry's own order parameter, namely Phonon Beam, Magnon Wave, Ferron Switch, Electromagnon Drive and Higgs Oscillation

Two attacks are carried by a quasiparticle with no charge, no spin and no order parameter of its own, so no cloud touches them at all: Majorana Split and Vison Loop. Landau's and Skłodowska-Curie's tunable moves are screened as whichever quasiparticle you have tuned them to, so an untuned one counts as a phonon.

Picking a move picks the cloud. Only one can be active at a time; switching means talking to Kondo again, so holding one quantum number always means giving up the other two.


Figure 35: Kondo's self-buff shop: his three screening moves, with the one currently being held marked active

Each of these three can also be leveled up at Feynman's, same as any other move, leveling one deepens its screening past the base half, up to a cap so it never becomes complete immunity.

### Franklin's Scatterings

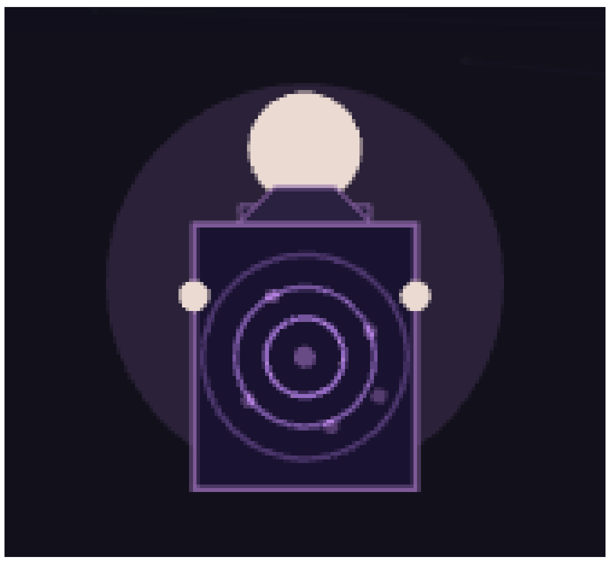

Figure 36: Franklin: a figure holding up a sheet of film carrying diffraction rings and their scattered spots

Teaches three passive abilities, always-on effects for the whole battle, not moves you pick each turn. You can learn all three, but only one is ever equipped at a time; talk to Franklin again to switch. Themed around X-ray diffraction of a defect-riddled or porous crystal, the way a real diffraction pattern blurs from sharp spots into diffuse rings as a sample gets more disordered.


Figure 37: Franklin's panel: your crystal wearing the halo of whichever passive is active, beside the three passives and their prices

**Franklin's passives**

| Passive | Effect | Cost |
|---|---|---|
| Diffraction Shadow | A defect-riddled lattice scatters and attenuates an incoming blow, the way porous carbon attenuates an X-ray beam. | 40 |
| Satellite Reflection | A critical hit throws off a secondary diffraction peak: a bonus follow-up damage tick. | 45 |
| Amorphous Halo | A diffuse, defect-broadened halo softens the quasiparticle-mismatch double damage to a smaller multiplier. | 45 |

**Skłodowska-Curie's Experiments**

Figure 38: Skłodowska-Curie: a ray-crowned spire figure with a glowing core, ringed by radiating sparks

Guards World 10 and is regarded as the leader of the guardians' circle. Sells two quiz-gated Ultimate moves, "[Quasiparticle] Meteor" and "[Quasiparticle] Nova", each far more powerful than any ordinary move in the game, and gated to match: landing one takes three physics questions in a row, all correct, drawn from a broad pool spanning everything the course has covered rather than any one world's topic. Miss even one question and the move whiffs for zero damage, though your turn is still spent.

Picking which quasiparticle an Ultimate move carries isn't a flat purchase like it is everywhere else: each quasiparticle class costs 1000 qumatessence to unlock *per move*, after which retuning back to that class is free forever. Land a full 3-for-3 hit and it plays the longest, most dramatic summoning animation in the game.


Figure 39: Skłodowska-Curie's panel: Phonon Meteor striking its stage mid-summoning, its phonon class already unlocked and free to carry, other classes at 1000 qumatessence each